



Hana SOUMMAR

Junior Data Scientist

Email: soummarhana@gmail.com
Mobile: +212 630474703
LinkedIn : Hana Soummar

A Master's graduate in Information Systems and Intelligent Systems from **INSEA**, actively seeking a full-time position in **artificial intelligence** or **data science**. Rigorous, adaptable, and proactive, I bring a solid foundation in AI and data science, along with industry experience, to contribute effectively to real-world and impactful projects.

EXPERIENCE

- **Oracle R&D Center - Research Assistant** **February 2025 - August 2025 (6 months)**
 - Developed an **AI-powered Support Ticket Similarity Matching system** leveraging advanced NLP and **Deep Sentence Embeddings** to automatically identify and group semantically similar customer support tickets.
 - Designed an end-to-end semantic retrieval pipeline combining vector similarity search in **Chroma DB (Vector Database)** and a **Cross-Encoder Reranking Model** to improve matching accuracy.
 - Built a scalable full-stack architecture with **FastAPI** and **React.js**, enabling real-time semantic search and analysis.
- **3D Smart Factory - AI Intern** **July 2024 - September 2024 (3 months)**
 - Designed a **3D Indoor Scene Segmentation System** using point cloud processing to automate spatial understanding of complex indoor environments.
 - Developed deep learning models using **PyTorch**, applying advanced 3D semantic segmentation techniques.
 - Integrated **Weights & Biases (W&B)** and **Open3D** to optimize experimentation and visualization workflows.
- **Vala Orange – Web Development Intern** **April 2023 - June 2023 (3 months)**
 - Developed a full-stack Human Resource Management system using **Laravel**, featuring interactive dashboards for real-time operational monitoring.

EDUCATION

- **National Institute of Statistics and Applied Economics (INSEA)** **2023 – 2025**
 - Master in Information Systems and Intelligent Systems (M2SI)
- **Faculty of Science and Technology (FST)** **2020 – 2023**
 - Bachelor Degree In Computer Science - (*valedictorian*)

PROJECTS

- **Moroccan Darija NLP Suite – Research & Fine-Tuning**
 - **Fine-tuned MarianMT** on a curated corpus of **44,714 parallel sentences** to enable high-quality translation of informal Moroccan Arabic into English.
 - **Built and cleaned** a custom domain-specific dataset to adapt **DarijaBERT** using **LoRA (Low-Rank Adaptation)**, achieving robust classification across health, economy, and education.
 - **Engineered** specialized preprocessing pipelines to standardize non-standardized scripts and address critical **low-resource NLP challenges** for dialectal Arabic.
- **LinkedIn Post Generator Using Large Language Models**
 - Developed an **LLM-powered content generation tool** capable of analyzing writing style and generating tailored LinkedIn posts.
 - Built an interactive application using **Streamlit**, integrating **Llama 3.3** and **LangChain** via the **Groq API**.

SKILLS

- **Programming Languages:** Python, R, Java, C, Bash, PL/SQL
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn
- **NLP & Generative AI:** Hugging Face Transformers, Sentence Transformers, Fine-Tuning, LangChain
- **Data Engineering & Databases:** PostgreSQL, Oracle, MongoDB, Chroma DB (Vector Database), Airflow, Talend
- **Backend & APIs:** FastAPI, REST APIs
- **Web Scraping:** Selenium, BeautifulSoup
- **Tools & Cloud:** Git, Docker, OCI (Oracle Cloud Infrastructure), Power BI
- **Languages:** Arabic (Native), Tamazight (Native), French (Fluent), English (Professional)